

Research Programs in Catalysis: Prior Work, Original Questions, and Experimental Decisions

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Overview and recommended priorities

This portfolio recommends developing **four ideas in the main document**, in the order shown below. Each connects an important catalytic limitation to a bounded intervention, a fair material or operating comparator, and a prediction that can be tested on independent experiments. Four is an editorial choice: it gives the two strongest development candidates full treatment while retaining two more immediately bounded studies with concrete resource-use decisions. The remaining ideas are recorded briefly in Section 5; their present limitations do not justify equivalent experimental scope.

Rank	Program	Contribution sought and reason for its position
1	Physical water management (Section 1)	Test protection by late hydrophobic-polymer addition at one catalyst state and water challenge, then conditionally predict useful output during a nearby reversible humidity pulse. A substantial published formulation effect motivates the work; the pulse prediction does not establish a damage law across histories.
2	Cyclic oxide steam compatibility (Section 2)	Determine when steam changes selective oxygen delivery and whether protection outperforms recovery. This addresses a specific gap between an Ar-purged durability experiment and a steam-based process proposal.
3	Polymer ethenolysis catalyst demand (Section 3)	Predict whether ethylene operation can replace some fresh Na/alumina during reuse while preserving polymer-derived output. The resource decision is concrete, although rescue and pressure effects have close prior art.
4	Promoted Ag retention (Section 4)	Predict when Ni adds useful ethylene oxide output beyond established chloride policies. Historical Ni-retention evidence makes this a demanding, tightly bounded material/operation study.

This ordering concerns scientific development and space in the manuscript, not a commitment to run four campaigns simultaneously. Water management is the first feasibility priority and cyclic oxides the second feasibility priority. Polymer and Ag may offer the best first experiment where their reactors, reference materials and analytical methods already exist. Laboratory access, preparation reproducibility, achievable uncertainty and actual costs have not been established. None of the four currently warrants an unrestricted campaign or a claim of competitive industrial lifetime.

The programs preserve different decisive comparisons. Water management introduces the additive

after common catalyst conditioning and tests function as well as water response. Cyclic oxides first compare ordinary steam and Ar operation before using equal-dose CO₂ timing and a common analytical reset to investigate a consequence. Polymer work crosses the selected ethylene condition with fresh-Na dose instead of comparing one low-pressure/low-dose choice against an incompletely optimized reference. Ag compares untreated, sham-processed and Ni-treated materials, then the useful ordinary and recovery policies available to both Ni-containing and Ni-free choices. A simpler credible operating alternative is part of each material decision.

Three kinds of statement are kept separate throughout. Published observations are attributed to their sources and retain their preparation, measurement and time boundaries. Calculations are deductions from explicit assumptions or identified published data; they are not new experimental measurements. Proposed experiments specify the information needed for a decision, possible outcomes and limits of interpretation. Their contribution is a testable research design, not advance confirmation of a mechanism. In particular, the document does not establish lower local water exposure at Co sites, steam-induced Li loss, selective loss of polymer initiation, or preservation of a particular Ag–Re oxygen-transfer state.

The common standard for expansion is a consequential prediction: measured behavior should identify a material or operating choice and correctly anticipate its response under a withheld condition or history. A finite, well-resolved null can close that choice within its tested scope. More characterization is justified when it separates explanations that would lead to different decisions. Full elapsed time, inventories, product origin where relevant, and treatment burdens remain part of the comparison even when a mechanistic interpretation is successful.

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1 Physical water management: when does a neighboring hydrophobic material protect a catalyst?

Proposed contribution. Determine whether a small polydivinylbenzene (PDVB) addition protects an already conditioned cobalt catalyst against a subsequent water disturbance, and establish whether measured water and catalytic responses predict useful output under a new disturbance history. The outcome would be a controlled test at a defined catalyst state and disturbance, with a conditional predictive rule within the calibrated range. A unique microscopic explanation of active-site drying is a conditional extension. No experiments proposed here have been performed.

1.1 Prior work, importance, and the remaining question

Physical hydrophobic promotion is established. Fang and colleagues demonstrated PDVB promotion of Co–Mn carbide for syngas conversion, including mixing-distance and water-cofeed effects; after removing PDVB from the used mixture, the recovered carbide performed comparably to the original catalyst. Li and colleagues extended physical promotion to copper-catalyzed methanol synthesis.[1, 2] These precedents rule out claiming physical mixing, post-use additive removal, cross-reaction transfer, or a contact-angle correlation as the original contribution.

The immediate platform is Fang and colleagues’ Co/SiO₂ study of cobalt Fischer–Tropsch synthesis, which converts CO and hydrogen (syngas) to hydrocarbons while producing water. C₅₊ denotes hydrocarbons containing at least five carbon atoms. With 0.025 g PDVB per 0.5 g catalyst, separately prepared 40–60 mesh granules maintained approximately 44% CO conversion and 88–90% C₅₊ selectivity over 1200 h at 220 °C and 2 MPa. The quartz-diluted reference initially converted more CO but declined strongly; nominal bed volumes were matched. Their water-cofeed, heat-transfer, material-state, and mixing controls support a substantial functional effect.[3] The mechanistic interpretation nevertheless requires separating changes in retained water, water transport, reaction rate, and catalyst history.

For perspective, let X_{CO} be the fraction of inlet CO converted and $S_{\text{C5+}}$ the published carbon-based selectivity, counting carbon atoms rather than product molecules. Integrating their published series over the common 25–585 h interval gives

$$Q^* = \int X_{\text{CO}}(t)S_{\text{C5+}}(t) dt = 116.0 \text{ h (reference), } 219.8 \text{ h (PDVB)}. \quad (1)$$

This secondary calculation uses Fang’s Source Data workbook, sheet “Figure 1,” with equal nominal CO feed. Conversion and selectivity are separately interpolated piecewise linearly, then their product is integrated on the combined time grid over 25–585 h. The output-proxy ratio is 1.89, or 1.80 after charging the additional 5% polymer mass; the latter excludes quartz mass. It is not a complete carbon balance, an uncertainty interval, or a lifetime extrapolation. Early operation before 25 h is excluded. The practical question is therefore credible, but superiority to an optimized cobalt formulation or regeneration policy has not been established.[3]

Water history and pore environment also have direct precedents. Hanssen and colleagues already compared water pretreatment with subsequent cofeeding and a dry return on alumina-supported cobalt.[4] Wolf and colleagues used externally supplied water at very low conversion followed by a dry return to connect cobalt oxidation, sintering, and lost function on graphite. Paterson and colleagues showed that activation-water exposure affects subsequent cobalt performance.[5, 6] Yang and colleagues related silica-shell geometry to deactivation and distinguished water-related and heavy-hydrocarbon-related behavior.[7] More recently, Sengupta and colleagues reported that strongly

retained oxygenates change catalyst water affinity during operation.[8] Thus a dry fresh-material contact angle cannot be assumed to represent a working catalyst. The proposed distinction is *late addition after a common conditioning protocol*, followed by a controlled functional comparison and a prediction outside the calibration history.

1.2 What a faster water transient can establish

At equilibrium, surfaces can hold different amounts of water at the same chemical potential. Hydrophobicity alone cannot maintain a chemical-potential gradient across an otherwise equilibrated connected gas. Reaction, imposed flow, or another source or sink can maintain a gradient; wettability can then affect storage, liquid configuration, and the resistance to transport. This distinction preserves the reported promotion while rejecting equilibrium loading differences as a sufficient explanation of continuous drying.

A minimal reversible-storage balance makes the ambiguity explicit:

$$B \frac{dc}{dt} = q - G(c - c_b), \quad \tau = \frac{B}{G}, \quad c_{ss} - c_b = \frac{q}{G}. \quad (2)$$

Here c and c_b are local and boundary concentrations in a common gas-referenced coordinate, B is differential storage capacity (volume), G is exchange conductance (volume/time), and q is water production (amount/time). The expressions assume a locally linear, time-invariant compartment with a specified source. Decreasing B shortens both filling and emptying without lowering the stationary source-generated excess. It can increase exposure earlier in an upward step while shortening the subsequent tail. Consequently, less storage need not improve durability. Increasing G also shortens the transient but lowers the steady excess. The reported syngas-to-Ar decay times of approximately 80 and 175 s cannot distinguish these alternatives: that switch also changes production and adsorbed species.[3]

Even absolute inventory plus several outlet transients need not identify the producing region. For two pools with concentrations c_i , capacities B_i , and conductances $G_i = \lambda B_i$, λ is their common exchange rate constant (inverse time). Let $B_T = B_1 + B_2$ and $W = B_1 c_1 + B_2 c_2$. With total water production q_T , inlet concentration c_{in} , a well-mixed gas volume V at concentration c_g , and actual volumetric flow F ,

$$\dot{W} = \lambda(B_T c_g - W) + q_T, \quad (3)$$

$$V \dot{c}_g = F(c_{in} - c_g) + \lambda(W - B_T c_g). \quad (4)$$

Only total capacity and total source enter the measured outlet c_g . If all production is in pool 1, its steady pool-to-gas excess is $c_{1,ss} - c_{g,ss} = q_T/(\lambda B_1)$. Changing B_1 from $B_T/4$ to $3B_T/4$ reduces this excess threefold without changing any outlet response at identical aggregate initial conditions. Since $c_{g,ss} - c_{in} = q_T/F$, the total pool-to-inlet excess includes an unchanged q_T/F term and does not generally scale threefold. This constructed example is not a model fitted to PDVB; it shows why a source-specific or spatial constraint is needed for an active-site claim. Isotope signals do not automatically supply that constraint.

The program therefore separates three levels of evidence: measured formulation response, prospective prediction of catalytic function, and identification of a local mechanism. A correct outlet fit meets only the first. A stationary transport explanation must also constrain the location of the source and relevant resistance. For example, changing an external resistance cannot remove an unchanged internal resistance. Changing granule size alone does not resolve their relative contributions.

1.3 Experiment 1: qualify the platform and the intervention

Reproduce the low-dose, separate-granule contrast with measured catalyst inventory, bed volume, temperature, pressure drop, carbon balance, and useful-product rate. Keep the source’s distinct configurations separate: its imaging of a powder co-granulate and some high-dose controls do not define the low-dose longevity packing. The published restart procedure is incomplete: synthesis includes reduction and passivation, whereas Supplementary Note 10 calls the standard test hydrogen-pre-reduced without fully specifying that restart. Establish and document one reproducible activation sequence before interpreting the new comparison; do not silently import the 300 °C analytical pretreatment.[3]

The late intervention requires cooling and repacking. Use independently preweighed catalyst charges from the same preparation, each with identical initial quartz dilution that remains with the charge throughout handling. Condition without PDVB under syngas, initially using the platform’s H₂/CO/Ar ratio of 64/32/4 at 220 °C and 2 MPa. The parent pilot selects a fixed duration beyond startup, using rate and selectivity drift over a defined observation window with tolerances set from pilot variability. Freeze the duration and complete feed, flow, temperature, and pressure schedule before randomization; report pre-handling rate/selectivity drift and retained-liquid observations. This defines an operationally conditioned age, not microscopic stationarity or cessation of changes in water affinity.

Randomize charges within preparation and run-order blocks to the four conditions in Table 1. Common protocol controls intended history; independent repetitions quantify the remaining state variability. Recover each complete catalyst-and-quartz zone under inert gas, preserve retained liquid, and use the original dry catalyst inventory as the mass denominator after demonstrating quantitative solid recovery. Record liquid loss separately. All charges undergo identical cooling, transfer, and restart; no solvent washing or new reduction is introduced between conditioning and challenge.

Table 1: The decisive late-additive comparison. Each condition has independent catalyst charges; repeated chromatograms are not independent replicates.

After common conditioning	Water challenge	Time-matched control
0.05 g PDVB/g catalyst	Low–high–low water	Low water throughout
Matched quartz diluent	Low–high–low water	Low water throughout

Repack each whole recovered charge with either 0.05 g PDVB/g catalyst or quartz occupying the same measured bulk volume, to equal final bed volumes; no initial diluent is selectively removed. Keep catalyst granule size fixed and measure actual void volume and pressure drop rather than assuming equal hydrodynamics. Include the initial and final packing, heat and flow behavior, and solid/liquid recovery in the handling qualification. This pilot compares restarted parent charges with an uninterrupted parent control. A reproducible shift can define the post-handling starting state; uncontrolled solid/liquid loss or erratic restart closes the late-addition route. Handling cannot recreate uninterrupted industrial operation, and this limitation remains part of the interpretation.

1.4 Experiment 2: distinguish protection from an activity penalty

Use one imposed-water challenge selected in a parent-only feasibility study. A starting design is 220 °C, 20 bar total pressure, inlet $p_{\text{H}_2} = 10$ bar and $p_{\text{CO}} = 5$ bar, with water raised from 0.5 to 4 bar

by replacing Ar at fixed inlet molar flow. These are design choices, not a demonstrated damage threshold for Co/SiO₂. Select the plateau and recovery durations before the four-condition comparison: the plateau must outlast measured equilibration and leave measurable but noncatastrophic residual loss in the parent. If no suitable disturbance exists within the qualified operating range, the proposed damage test has failed qualification.

Target at most 2% CO conversion in the more active arm. Assuming one water per CO converted, the generated/supplied molar-water ratio is

$$\frac{\dot{n}_{\text{water,gen}}}{\dot{n}_{\text{water,in}}} \simeq \frac{X_{\text{CO}} y_{\text{CO,in}}}{y_{\text{water,in}}} = 0.025 \text{ at 4 bar water,} \quad 0.20 \text{ at 0.5 bar water.} \quad (5)$$

Thus external water dominates the high plateau much more strongly than recovery. Determine actual ratios from balances, including CO₂, oxygenates, gas contraction, and unresolved oxygen storage. Low conversion suppresses whole-bed reactant changes; it does not prove small intragranular gradients. Before fixing the flow, verify that a modest further flow increase does not materially change mass-normalized diagnostic rate. An order-of-magnitude planning calculation, retaining the source’s nominal promoted CO consumption, requires about 68 L (g catalyst h)^{−1} to reach 2% conversion at inlet CO fraction 0.25. For a 50 mg charge this corresponds to about 3.4 standard L/h of gas and 0.55 g/h supplied water at 20% steam (0 °C volume convention). These are equipment-sizing estimates, not rate predictions. Reducing inventory can make the test practical without reducing production per individual granule.

For each charge measure pre-challenge and post-recovery C₅₊ carbon-production rates, r_{pre} and r_{post} , after defined settling windows. Let $R = r_{\text{post}}/r_{\text{pre}}$. The primary protection contrast is the interaction

$$D = (R_{\text{PDVB,challenger}} - R_{\text{PDVB,sham}}) - (R_{\text{quartz,challenger}} - R_{\text{quartz,sham}}). \quad (6)$$

A positive D means that PDVB reduces the fractional functional penalty associated with the challenge beyond time-matched drift. Report both pre-challenge and post-recovery absolute rates for all four conditions alongside D ; a favorable fraction can conceal an inferior catalyst bed. The second decision is the direct difference in integrated C₅₊ carbon output per complete bed volume over the same predefined restart–challenge–recovery horizon. That comparison charges the initial activity penalty and includes startup product inventories and collection delays. Report catalyst-plus-additive mass productivity separately. The first endpoint tests protection, and the second tests whether protection is useful on that horizon.

Choose the number of independent blocks from pilot variability and a predeclared minimum consequential difference in retention and output; no fixed percentage is justified before that information exists. Report intervals incorporating preparation and run variability, calibration, and balance uncertainty. A null is informative only if its interval excludes the consequential benefit. PDVB remains present during recovery, so improved retention demonstrates the function of the complete formulation, not the number of surviving cobalt sites or permanent irreversibility.

A positive contrast after late addition excludes an explanation confined entirely to the original conditioning, but it leaves effects during restart, the water transient, and ongoing liquid or chemical changes open. Any parent checks used to reject a whole-bed reaction-water source explanation must bracket the measured formulation-dependent water histories during both challenge and recovery, including their uncertainty. Plateau-only bracketing cannot exclude that explanation for post-recovery retention: generated water has a larger relative contribution during recovery. If the relevant histories cannot be bounded, retain the source explanation while reporting the formulation contrast.

A null, after successful reproduction and qualification, limits the supported benefit of late addition in this specific state; it does not establish that startup protection caused the published result.

1.5 Experiment 3: predict one useful disturbance response

Pursue a predictive branch only if the preceding comparison or reversible water response supplies a resolvable functional question. Zero-length-column work shows how independent storage information and flow-dependent transients can constrain water exchange in silica gel, but its passive, low-temperature coefficients cannot be transferred to a reacting, liquid-bearing cobalt bed.[9]

First qualify water-specific blanks at the actual temperature, pressure, and composition, including heated transfer lines, pressure reduction, and detector memory. An inert tracer establishes gas delay but cannot correct water-specific storage. On representative conditioned parent and PDVB packings, apply small upward and downward humidity steps at two flows with unchanged inlet H_2/CO partial pressures, a reduced-amplitude repeat, and baseline returns. Integrate absolute water rates; do not normalize away the inventory information. Measure product rates fast enough to bound the changing chemical water source. A total-mass uptake decrease must also be corrected for dilution: for additive/catalyst mass ratio a , noninteracting uptake is $(u_{cat} + au_{add})/(1 + a)$ even with unchanged catalyst loading.

Fit only the simplest response representation consistent with both step directions, flows, source balance, and independent packings. In a reacting compartment, linearization of Eq. 2 with $q = q(c)$ gives a relaxation rate $(G - \partial q/\partial c)/B$. This decay expression assumes $B > 0$ and a stable stationary state with $G > \partial q/\partial c$, the derivative evaluated at that state; a measured pole is not automatically G/B . If source feedback, blank uncertainty, or changing liquid state prevents a meaningful coefficient, stop the conductance claim. Empirical input-output prediction can still be tested, but its coefficients must not be relabeled local transport properties.

The concrete validation is a partial-duration water pulse withheld from fitting. Preselect a fixed decision horizon from pulse onset, representing the available production window, and an acceptable C_{5+} carbon-output deficit relative to undisturbed operation. This endpoint informs whether to tolerate a humidity excursion of the proposed duration within that output budget. Calibrate each formulation’s nearby reversible product response independently and account for product-detector and liquid-collection delays. Select a pulse duration for which surviving models predict consequentially different output deficits over that fixed horizon, and freeze predictions before collecting it. Distinct water traces alone do not qualify: different time constants with equal steady gains can yield the same deficit integrated through complete recovery.

Predict both the water trace and the selected functional endpoint, with an endpoint prediction interval narrower than the predeclared consequential contrast. Compare with a simple baseline using inlet humidity, the calibrated steady product response, and independently measured apparatus delays; this uses the same calibration and validation data. Validate on independent charges without refitting parameters to the withheld pulse. Passing this test supports a bounded operating-history rule over the calibrated state and amplitude range. If transient characterization does not improve on the baseline, claim only the supported empirical rule. Neither outcome validates a damage law at much higher water exposure.

1.6 Decision and scope of the original contribution

The program has two credible contributions: a controlled test of late low-dose PDVB at a defined catalyst state and disturbance, and, conditionally, a prospective rule connecting a measured disturbance to useful catalytic output within the calibrated range. Their importance is deciding whether a simple formulation change pays for its initial penalty under the tested conditions and when faster water clearance is an inadequate selection criterion. A mechanism paper would require an additional selective constraint on the producing region, shown to remain valid when the additive changes liquid and catalyst state. Outlet transients alone cannot provide that guarantee.

Close the affected branch if the source effect cannot be reproduced, handling destroys interpretability, the challenge has no measurable dynamic range, or the predicted functional difference is below measurement resolution. If fractional protection is real but cumulative output is worse, report that tradeoff and do not claim a practical advance. A useful formulation result can survive failure to identify the stationary mechanism. Conversely, a well-fitted water curve without a consequential functional prediction does not justify an expanded materials or isotope campaign. Commercial claims require a subsequent complete-cycle comparison with an optimized catalyst, including regeneration, additive inventory, bed volume, and replacement time.

2 Cyclic oxides: steam compatibility and recovery of selective oxygen delivery

Proposed contribution. Test whether replacing the post-oxidation Ar purge with a selected steam-rich purge changes useful function in Li_2CO_3 -coated $\text{La}_{0.8}\text{Sr}_{0.2}\text{FeO}_3$ (LSF) under ordinary cycling without supplementary CO_2 , then test recoverability and whether concurrent CO_2 is more useful than delayed treatment. The prospective result is a compatibility and recovery comparison under specified histories, with a conditional prediction of complete-cycle ethylene output. Steam damage, lithium escape, and the benefit of protective CO_2 remain hypotheses. The post-ethane purge remains dry Ar and is outside the substitution claim.

2.1 Why the question matters, and what prior work already establishes

Ethane oxidative dehydrogenation produces ethylene while removing hydrogen as water. In chemical looping, an oxygen carrier supplies oxygen during the ethane step and is subsequently reoxidized, separating ethane conversion from the gaseous oxidant feed. The ideal selective reduction is



The challenge is delivering enough oxygen for useful conversion without oxidizing hydrocarbons to CO or CO_2 . Gao and colleagues established a carbonate-coated LSF carrier and attributed its selectivity to a molten layer that blocks nonselective sites and transports peroxide species associated with $\text{Fe}^{4+}/\text{Fe}^{3+}$ redox. They report approximately 0.42 wt% oxygen consumption over the first six minutes at 700 °C. The supported coating is reported to melt below bulk Li_2CO_3 ; hydroxide formation need not be the event that first makes it liquid.[10]

Brody and colleagues subsequently cycled 300 g of coated LSF for more than 1200 elapsed hours. During the first 1000 h at 700 °C, each cycle contained one minute of ethane feed, five minutes Ar purge, five minutes oxidation, and five minutes Ar purge. Average C_{2+} carbon yield declined from 53.2% over hours 0–300 to 47.24% over hours 600–1000. C_{2+} means ethylene plus C_{3+} products, not ethylene alone. Later temperature and flow changes improved performance, so the later optimum was not sustained throughout the whole test. The methods specify 10 wt% carbonate whereas the conclusion says 20 wt%; actual loading must be measured.[11]

The process model in that study replaces the experimental Ar purges with steam heat-exchange steps. It does not measure how steam changes the coating.[11] This gap is the strongest reason to investigate steam compatibility. Steam cannot explain the decline already observed under Ar; explaining that decline is a separate aging problem. Nor does the source establish a continuously producing, thermally self-sufficient plant simply by observing exothermic half-cycles.

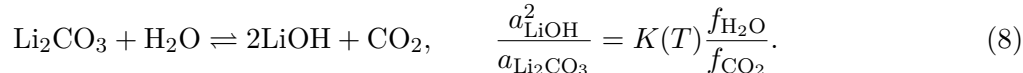
The chemical premise and much of the broader regeneration idea are also established. Barckholtz and colleagues observed carbonate/hydroxide responses to steam and CO_2 in a mixed Li/Na melt, although their Raman signals did not give absolute species concentrations.[12] Chacko’s dissertation reports actual mixed-carbonate/LSF cycling at 800 °C with 45 wt% salt and CO_2/O_2 regeneration. It includes carbonate decomposition during inert purges and H_2 evolution during regeneration, attributed to reduced iron reacting with water released during hydroxide carbonation. These are close precedents for coupled coating/carrier regeneration, despite the different formulation and temperature.[13, ch. 4.1]

Even prevention versus incomplete delayed recovery is prior art. Fereres and colleagues changed a bulk Li/Na/K carbonate melt from air to CO_2 after 30 days at 640 °C: further mass loss stopped

without restoring the initial state under their treatment.[14] The proposed originality is therefore specific: test the post-oxidation steam substitution and its recoverability in this lower-loading coating, assess whether a measured state predicts a new operating history, and determine whether an intervention repays its gas and time burden. Another demonstration that CO₂ stabilizes carbonate would be insufficient.

2.2 The chemical hypothesis and its competing explanations

A candidate component equilibrium is



Here a denotes condensed-phase activity and f gas fugacity; the fugacity ratio approximates the partial-pressure ratio in the proposed atmospheric-pressure diagnostic. Lower steam/CO₂ ratio favors carbonate relative to hydroxide at fixed temperature. This direction does not supply a composition, equilibration time, or lithium-loss rate for supported LSF. Nonideal melt chemistry, substrate interactions, and changing coverage preclude importing a fuel-cell equilibrium constant uncritically. Barckholtz’s experiments used hours per condition; they do not demonstrate a five-minute coating response.[12]

Several outcomes could follow. A retained salt might change reversibly between chemical forms; it might redistribute away from important interfaces; lithium might move onto reactor contact materials or leave in transported material; or the carrier could change redox state or phase. These effects can occur sequentially. Lithium found downstream does not identify LiOH vapor, and unchanged total lithium does not establish unchanged coverage. A carbonate surface-signal decrease is not a quantitative lithium-loss measurement.

Restoration is also chemically coupled. The balanced reverse reaction,



releases water even with a dry inlet. Chacko’s printed analogous reaction omits the factor of two before hydroxide; Eq. 9 corrects that stoichiometry. The accompanying water can participate in carrier oxidation.[13, ch. 4.1] Thus equal fed CO₂ or equal mean Fe valence cannot establish equivalent coating–carrier interfaces. The timing comparison measures a total treatment-order effect and restricts its interpretation using inventories and intermediate states.

2.3 Experiment 1: ordinary-cycle compatibility, then a timing comparison

Table 2 separates the operating comparison from the diagnostic and policy sequences. Ordinary oxidation remains part of every cycle; a special reset is an additional treatment.

Prepare one reproducible coating composition, quantify its initial Li content and phases, and establish stable short-cycle function with a dry Ar protocol. Use the methods-based nominal 10 wt% loading only as the starting recipe, documenting the measured final composition and loading convention. Begin the intervention immediately after ordinary oxidative regeneration. This selects the relatively oxidized post-oxidation purge; it does not yet represent the reducing post-ethane purge.

First compare ordinary cycling with a steam-rich post-oxidation purge against ordinary cycling with Ar, without supplementary CO₂ in either arm. Use 700 °C and atmospheric pressure, with the

Table 2: Sequence map for the proposed cyclic-oxide comparisons.

Sequence	Exposure	Special-reset timing	Primary endpoint
Ordinary operation	Five-minute post-oxidation steam-rich or Ar purge; fixed cycle block	None during the block; one terminal reset after operating output is measured	Pre-reset cycle and cumulative ethylene; separate post-reset recovery assay
Five-arm timing diagnostic	Two five-minute segments: wet/dry early/late CO ₂ , plus Ar reference	One common reset after exposure, or after the fixed accumulation block	Post-reset ethylene and timing contrasts; intermediate state samples support attribution
Usable policy blocks	Concurrent protection, delayed restoration, untreated steam and Ar at the selected duty	Managed policies: frozen treatment cadence. Untreated/Ar: none. Terminal analytical reset follows counted output	Ethylene per complete time and resource demand; terminal function reported separately

source’s one-minute ethane, five-minute Ar, five-minute oxidation, and five-minute final purge as the starting schedule. Select one steam fraction in Ar, flow, and temperature trajectory from equipment qualification and the intended heat-exchange duty; hold the other cycle conditions fixed. Freeze that composition, the initial conditioning, cycle count, and functional averaging windows before comparing independent charges randomized within preparation and run-order blocks. Choose cycle and replicate counts from pilot variability, method uncertainty, and a predeclared consequential output loss, not by cycling until failure appears. Ordinary oxidation occurs every cycle; no special carbonation reset occurs during this block.

Measure cycle-resolved ethylene and cumulative output throughout, including a fixed terminal window *before any special reset*. Report the steam-minus-Ar output difference and its uncertainty on equal initial mass and feed bases. This is operating performance, including any steam carryover or reversible inhibition, not proof of permanent damage. Do not insert an extra dry conditioning step before measuring it. Then apply the pilot-qualified terminal reset below once to each completed steam and Ar history and measure a common dry-cycle functional block. Compare the pre-reset and post-reset outputs with their matched Ar history: a steam penalty before treatment followed by recovery is distinct from tolerance during ordinary cycling. Recovery means performance under that terminal treatment, not proof of a unique repair mechanism. The pre-reset blocks also supply the untreated policy comparator for Experiment 2.

If a functional or material contrast warrants diagnosis, use the two five-minute segments in Table 3 on separate specimens at 700 °C and atmospheric pressure. Ten mol% steam and 5 mol% added CO₂ are initial diagnostic settings, not an optimum or a simulation of nearly pure steam. Their results do not substitute for the ordinary-cycle comparison at the selected steam-rich duty. Keep total inlet molar flow, temperature history, pressure, specimen mass, and total elapsed exposure fixed. Calibrate switching holdup using an empty reactor and the actual water/CO₂ signals. Match the *integrated inlet* CO₂ dose within each timing pair and report measured background CO₂; equal setpoints alone are insufficient. Uptake and endogenous release remain outcomes rather than quantities forced to match.

The two dry timing arms are necessary because early CO₂ could protect against a dry exposure independently of steam. After the common reset described below, measure the mean ethylene amount per cycle and initial catalyst mass, q , over a fixed block for each arm. Define the steam-dependent

Table 3: Timing and steam are crossed to distinguish a steam-dependent timing effect from ordinary dry-gas history. All unspecified gas is Ar.

Arm	First five minutes	Second five minutes
Wet, concurrent (P)	10% steam + 5% CO ₂	Dry Ar
Wet, delayed (R)	10% steam	5% CO ₂ , dry
Dry, early CO ₂	5% CO ₂ , dry	Dry Ar
Dry, late CO ₂	Dry Ar	5% CO ₂ , dry
Ar reference	Dry Ar	Dry Ar

timing contrast as

$$D_{\text{steam}} = (q_{\text{wet,P}} - q_{\text{wet,R}}) - (q_{\text{dry,early}} - q_{\text{dry,late}}). \quad (10)$$

A positive wet P–R difference establishes a timing benefit under steam; positive D_{steam} adds evidence that steam changes that benefit. Neither alone proves hydrolysis or lithium volatilization. Also report $q_{\text{wet,P}} - q_{\text{dry,early}}$ and $q_{\text{wet,R}} - q_{\text{dry,late}}$, with uncertainty intervals against the predeclared consequential loss. These compare steam substitution within each CO₂ schedule. Report every arm’s absolute yield and its difference from Ar as an overall history comparison: both wet arms might be damaged equally even when their timing interaction vanishes. A null here concerns function after the specified CO₂ history and reset, not untreated steam compatibility.

Use independently run sister specimens, randomized within preparation and run-order blocks, for the first-segment endpoint, completed exposure, and final reset endpoint. At every endpoint, use in situ observation or a validated termination, quench, and controlled transfer, documenting the elapsed time and gas history. The first-segment samples can locate changes already present during steam only for observables shown to survive sampling. Withhold temporal species or spatial attribution if termination changes the measured state. Final specimens alone cannot distinguish steam damage from changes during the later Ar/CO₂ segment. Repeated cycles on one specimen do not replace independent runs. Determine the replicate count from pilot variance and a predeclared minimum consequential yield or lithium-inventory difference.

2.4 A common reset, defensible inventories, and a persistent function assay

Use one common terminal carbonation/oxidation sequence for the recovery comparison and for all timing arms. It is additional to ordinary cycle oxidation and follows the ordinary comparison’s pre-reset output measurement. In a pilot, choose dry-inlet CO₂/O₂ conditions and a duration sufficient for the slowest responding specimen to reach stable net uptake and a reproducible carbonate-sensitive spectrum. Freeze that sequence for the comparison. A joint CO₂/O₂ endpoint limits a new decarbonation interval while reoxidizing the carrier; its adequacy must be demonstrated on this formulation. Measure CO₂, O₂, H₂O, CO, and H₂ throughout. Apply the same state-preservation qualification to phase and Fe-state measurements after the reset *and after the actual washout*.

The objective is a demonstrably completed treatment of the retained carbonatable fraction, not equal absolute carbonate inventory if Li has been lost. An uptake plateau can reflect slow transport or passivation, so it needs the independent species/state check. If a shared process-compatible reset cannot establish interpretable endpoints, report the resulting recovery limit without asserting absolute irreversibility. Any further reset needed to assess incomplete recovery must be applied comparably and charged to the recovery policy.

For the post-reset assay, verify CO_2 and O_2 washout before introducing ethane against thresholds qualified to exclude a consequential change in the first ethane pulse. Gao reports CO_2 inhibition on coated LSF in Fig. 4D; the explicit reversible switch trace in Fig. 4E is a separate molten- Li_2CO_3 /ethane/ O_2 experiment.[10] Residual gaseous O_2 can also change output independently of carrier oxygen. Establish the maximum washout needed across arms and apply the same interval in the diagnostic; then determine whether that interval alters the coating contrast. An effect erased by washout cannot be claimed as persistent protection. Follow identical ethane/oxidation cycles with dry Ar purges for a fixed block selected in the pilot to distinguish gas carryover from retained function.

Measure ethylene directly as the primary outcome, separately from the source’s C_{2+} objective, with conversion, total CO_x , other hydrocarbons, and retained carbon. Include carbonate-carbon capture, release, and inventory changes in the cycle carbon balance. Use these measurements to bound the carbonate contribution before assigning hydrocarbon-derived CO_x ; regeneration CO_2 alone is not a coke assay. If carbonate and coke contributions cannot be resolved, report total CO_x and the attribution limit rather than a precise combustion selectivity or coke amount. The close regeneration precedent also identifies this accounting problem.[13, ch. 4.1] No isotope campaign is required for the direct ethylene-output decision. Use a thermal blank to bound noncatalytic ethane conversion. Water and H_2 must be measured directly: carbonate/hydroxide and carrier oxygen changes make water alone an ambiguous lattice-oxygen reporter.

Close the Li inventory over active particles, grit, accessible reactor walls, collected particulates, and downstream condensates. Use complete digestion with blanks and recovery spikes for each matrix; establish sampling representativeness and extraction recovery before aging. Record each exposure segment’s capture separately, but recognize that earlier deposits may be released later. Bulk elemental assays locate total Li; ordinary EDX cannot establish Li coverage.

If redistribution controls the interpretation, a candidate is cross-sectional NanoSIMS mapping of Li-rich patches and rim-to-interior gradients, correlated with electron microscopy. Qualify submicrometre spatial resolution finer than the features being compared, matrix-dependent ion yields, sectioning artifacts, and beam-induced redistribution. Map intensities do not by themselves give absolute coating coverage, and unresolved films remain outside the claim. For carrier state, use phase characterization and, subject to access and qualification, ^{57}Fe Mössbauer spectra on state-preserved sister samples, following the method precedent in Gao.[10] Reference materials and replicate spectra must distinguish the proposed Fe-state or phase change above fitting uncertainty; overlapping components and bulk averaging can conceal an interfacial change. Carbonate-sensitive vibrational analysis supplies a separate chemical indicator. Instrument access and adequate resolution are dependencies, not established capabilities; close the corresponding mechanism claim if they cannot be qualified.

The analytical scale is demanding. On an explicitly assumed final-catalyst basis, 1 g containing 10 wt% Li_2CO_3 has about 18.8 mg Li. A 0.1% change is 18.8 μg , below a 1% bulk-inventory error. At 100 standard cm^3/min for five minutes, 5% CO_2 supplies approximately 1.12 mmol. Conversion of 0.1% of the assumed carbonate inventory corresponds to only 1.35 μmol CO_2 , about 61 ppm of the total gas dose against a 50 000 ppm feed (0 °C standard-volume convention). These are resolution examples, not predicted losses. Captures, absolute assays, and chemical signatures must complement subtraction of large inlet/outlet signals.

If a single diagnostic exposure is unresolved, choose the number of cycles from the full-method uncertainty budget, then accumulate captures over that fixed block. Include actual ethane and ordinary reoxidation intervals. Apply the assigned two-segment timing history once per cycle but

the special common reset only once, after the accumulation block; freeze and report this cadence before assigning a history-specific loss rate. This ten-minute diagnostic exposure remains distinct from the five-minute ordinary purge. Increasing temperature or extending steam exposure until damage appears would test a different condition, not establish failure of the proposed purge. A useful null excludes consequential change only under its stated gas, duty, cycle count, catalyst state, and recovery policy; it is not proof of lifetime steam tolerance.

2.5 Experiment 2: test a usable policy and its complete-cycle value

A consequential ordinary-cycle penalty, a recovery benefit, or a timing contrast justifies comparing management policies at the same selected steam-rich duty. Retain the post-ethane Ar purge. Freeze steam dose, CO₂ circulation limit, total flow, temperature trajectory, and treatment cadence before testing; any unavoidable change in steam delivery is part of the policy comparison. The diagnostic ratio cannot be transferred automatically: 10% steam/5% CO₂ gives a ratio of two; maintaining that ratio in a binary purge requires one-third CO₂. This is a composition calculation, not a required protective dose. At the same five-minute total purge flow and one-minute 81.6% ethane feed used in the source cycle, one-third CO₂ would circulate about 2.04 mol CO₂ per mol ethane fed in that one purge. Preserving steam dose instead of total flow increases gas throughput and requires a new heat and hydrodynamic comparison.

Compare concurrent protection, delayed restoration, untreated steam without supplementary CO₂, and dry Ar over independently replicated operating blocks. The pre-reset ordinary-cycle comparison supplies the latter two policies when conditions, block length, and preparation/run blocks match; otherwise include matched controls. Untreated steam and Ar receive only ordinary oxidation during the policy block. Fix any periodic CO₂ treatment in the managed policies from the pilot and record its cadence explicitly; a terminal analytical reset occurs only after the counted block and does not improve its reported output. Delayed restoration must be allowed a demonstrated process-compatible treatment, rather than intentionally insufficient recovery. If actual Li export is consequential and retained-material restoration fails, test one measured replenishment operation as a competing policy. Replacing Li can also change coverage, salt composition, and carrier structure, so it is not automatically a selective intervention on Li inventory. Count its interruption, heating, washout, and achieved recovery; do not infer an online replacement method from the original synthesis.

Let q_i be block-averaged desired product per cycle, τ the base cycle time, and h_i the average added treatment time, including periodic operations. For equal initial catalyst mass and feed pulses,

$$J_i = \frac{q_i}{\tau + h_i}, \quad J_P > J_R \iff h_P < \frac{q_P}{q_R}(\tau + h_R) - \tau. \quad (11)$$

Use integrated measured product over the entire block, including aging, rather than a recovered peak. As an illustration only, fully restoring the published Ar-aged C₂₊ yield from 47.24% to 53.2% permits at most

$$16 \left(\frac{0.532}{0.4724} - 1 \right) = 2.02 \text{ additional minutes per cycle} \quad (12)$$

before losing the throughput advantage, even before charging gas and heat. A five-minute added step every cycle needs a 31.25% output increase; the illustrated complete recovery provides 12.62%. This is a threshold for the documented Ar loss, not a prediction of steam damage or a full-life prevention benefit.^[11]

Report ethylene and C_{2+} productivity separately per initial catalyst inventory and elapsed time, together with net fresh and circulating CO_2 /steam, Li handling, and heat requirements per product. Recycling can reduce fresh CO_2 use while leaving circulation and separation burdens. A gas-intensive protection policy may remain a tradeoff, but a smaller Li signal alone cannot establish process value.

2.6 Experiment 3: predict recoverability under a new operating history

The scientific extension is to identify a measured state indicator that predicts persistent function under the specified recovery policy. If Li export is a leading candidate, calibrate per-cycle Li transfer for the protected and delayed policies and relate retained inventory to subsequent ethylene output on independently aged specimens. Test whether that relationship is single-valued within uncertainty. If equal retained Li gives different function, an inventory-only model fails; assess an independently measured coating-distribution or carrier-phase indicator rather than fitting a separate loss constant for each history. Successful prediction establishes usefulness over the tested range, not causal dominance of Li inventory over correlated coating or carrier changes.

Use continuous protected and continuous delayed operation as calibration histories, then withhold one alternating protection/recovery schedule for validation. Fix its period, operating-treatment cadence, prediction model, and uncertainty intervals before collecting it. Predict cumulative Li transfer and cycle-resolved useful output through the operating block, plus standardized function after its single terminal diagnostic reset. Do not insert that terminal reset between cycles unless it is explicitly part of the operating policy and charged there. An additive per-cycle loss model supplies a clear initial prediction; failure under alternation reveals history dependence that the simple inventory description misses. Add a measured state only if it explains and predicts that failure on a further independent block. The result must improve upon a carrier-state-only model and a simple cycle-count aging description sufficiently to change a recovery or purge decision. A good fit to the calibration histories is not that result.

This sequence can yield a substantial program if the selected steam duty changes useful function and a measured state predicts recovery. An ordinary-cycle steam-minus-Ar interval excluding consequential loss supports compatibility without supplementary CO_2 only for that composition, duty, cycle count, conditioning, and ordinary oxidation policy; it closes the need to claim protection against an undetected penalty in that test. A pre-reset loss followed by terminal recovery instead supports recoverability under that treatment and leaves its operating cost to Experiment 2. Close the claim of persistent concurrent-over-delayed benefit after the stated reset only when the uncertainty interval for the predeclared direct contrast $q_{wet,P} - q_{wet,R}$ excludes a consequential advantage. Use the interval for D_{steam} to assess whether steam modifies that timing benefit. Individual recovery bounds remain separate conclusions, and neither comparison establishes that CO_2 treatment is unnecessary. If protection preserves function but cannot repay its complete-cycle burden against untreated steam and Ar, retain the scientific limit and reject the practical intervention. No result here establishes post-ethane steam compatibility or lifetime tolerance. Explaining the original Ar-aged decline requires representative Ar-aged material and a separate recovery study; it must not be substituted silently for failure of the steam hypothesis.

3 Polymer ethenolysis: maintain useful output with less fresh catalyst

Proposed contribution. Determine whether an operating change can substitute for part of the fresh sodium-on-alumina addition needed when a supported sodium/tungsten catalyst mixture processes successive polyethylene charges. The target is a predicted reduction in fresh catalyst solids per accepted polymer-derived carbon output, at a specified product quality and complete cycle time. The first intervention is one lower ethylene partial pressure on an existing semibatch platform. Selective loss of an initiating function is one possible explanation of reuse behavior, not an established result or a prerequisite for a useful operating rule.

3.1 Prior work and the remaining opportunity

Isomerizing ethenolysis combines double-bond migration with metathesis using ethylene to convert unsaturated chains into shorter olefins, ultimately including propylene. A saturated polyethylene (PE) feed additionally needs a route that creates reactive unsaturation. Conk and colleagues established direct conversion of PE and polypropylene with WO_3/SiO_2 and $\text{Na}/\gamma\text{-Al}_2\text{O}_3$. W/silica supplies olefin metathesis and Na/alumina supplies olefin isomerization: double-bond migration and reaction with ethylene repeatedly shorten a reactive chain. These established tandem roles do not determine how saturated PE first becomes reactive or which function changes during reuse. Their standard 1 g PE experiment uses 0.4 g of each solid at 320 °C; high fresh-batch output and heterogeneous catalyst reuse are already demonstrated.[15]

The most important benchmark is their three-charge reuse experiment. The starting mixture contains 0.4 g Na/alumina and 0.4 g W/silica. In the supplemented sequence, another 0.4 g Na/alumina is added *only before charge 2*; charge 3 receives PE alone. The mixture is retained, not separated into spent and fresh catalyst components. The unsupplemented sequence loses output, while that single addition restores output over the reported sequence.[15, Fig. 4B] Fresh-Na rescue is therefore the comparator, not a discovery proposed here. Rescue does not uniquely demonstrate unchanged W function: added Na material can alter unsaturation supply, isomerization, activation, contact, and the balance among tandem functions.

Initiation and regeneration are also active prior territory. Conk’s publicly available dissertation abstract proposes transfer dehydrogenation with ethylene as hydrogen acceptor, rather than the original paper’s cracking interpretation, and describes modified catalysts and dimethyl-ether-based regeneration. The body remains embargoed, so these disclosures constrain originality without supplying a verified complete protocol or performance comparison.[16] Patent WO2026011187A2 discloses oxidative treatments and subsequent dimethyl ether treatment; its Figure 21 caption names a Na–Fe formulation while adjoining regeneration procedures are not unambiguously consistent about catalyst identity. The disclosure cannot be treated as a demonstrated regeneration benchmark for every unmodified Na/W mixture.[17]

Theory already connects ethylene and reactive-chain concentrations. Guironnet and Peters developed a chain-population model and described its numerical extension to changing ethylene concentration; their displayed solutions use fixed ethylene concentration. Chen, Sadow, and Peters’ final 2024 model resolves finite isomerization kinetics and predicts saturation and possible ethylene inhibition. It concerns a different, molecular catalyst system with initially unsaturated chains, omits deactivation and ethylene gradients, and restricts physical applicability to relatively slow isomerization. The quasi-equilibrium limit is a numerical check, not its general physical assumption. The model does

not predict that lowering ethylene improves aged Na/W.[18, 19]

Pressure reduction and catalyst-supply effects were investigated experimentally in Wang’s earlier tandem system. Its pressure comparisons also changed other conditions, and its materials differ from Na/W.[20, ch. 4] Thus neither a pressure optimum nor a population-balance model alone is original. The remaining question is whether a measured change in operating response with catalyst history can replace a consequential amount of fresh material, and whether that substitution can be predicted. Its importance is reducing catalyst preparation, solids handling, and waste while preserving actual recovery of polymer carbon. No economic superiority is established by three successful batches.

3.2 Define the decision using carbon origin and complete inventories

Gross propylene production is an inadequate recycling denominator because ethylene contributes carbon to the products. The source’s labeled-polymer experiment shows polymer participation under its tested fresh-batch conditions; it is not a universal carbon allocation for every reused inventory and pressure.[15, Fig. 4A] Define Q_{PE} as moles of carbon originating in supplied PE and recovered in the specified accepted product over an entire sequence. For a propylene objective, count polymer-origin carbon in propylene; quantify higher olefins separately rather than silently treating them as equivalent product. Define

$$D_{\text{Na}} = \frac{M_{\text{Na/alumina,fresh}}}{Q_{\text{PE}}}, \quad D_{\text{W}} = \frac{M_{\text{W/silica,fresh}}}{Q_{\text{PE}}}, \quad J_{\text{PE}} = \frac{Q_{\text{PE}}}{T_{\text{complete}}}. \quad (13)$$

These are catalyst-solid demands, not elemental Na or W demands. Report elemental inventories and retained/discarded solids separately. Complete time includes heating, reaction, cooling, gas handling, unloading or recharge, and any regeneration required by the policy.

For decisive carbon attribution, use representative sequences with enriched ethylene at the *same enrichment throughout conditioning and every charge*, paired with unlabeled reference sequences. This avoids assuming that a small labeled-PE fraction represents all chain populations. Replace the source’s methane internal standard with He, measured by calibrated GC with a thermal-conductivity detector and Ar carrier gas. The source’s flame-ionization detector cannot measure He; retain separately calibrated hydrocarbon detection, with synchronized sampling and tracer recovery checked against known gas flows. Qualify separation from hydrogen, response in the actual gas matrix, and analytical access before decisive runs, then requalify reference behavior after the tracer change. Quantify any initial catalyst carbon and other carbon inputs. If appreciable unclassified carbon remains, two-source attribution is invalid.

Where PE and ethylene are the only contributing carbon reservoirs and isotope discrimination is bounded, the polymer-origin fraction in product j is

$$f_{\text{PE},j} = \frac{x_E - x_j}{x_E - x_{\text{PE}}}, \quad Q_{\text{PE}} = \sum_j N_{C,j} n_j f_{\text{PE},j}, \quad (14)$$

where x denotes *carbon atom fraction* of ^{13}C , $N_{C,j}$ the number of carbon atoms per product molecule, and the sum covers accepted products collected over the sequence. Derive x_j from calibrated isotopologue distributions with natural-abundance and fragmentation corrections. The published singly labeled propylene abundance is not itself a carbon atom fraction. Validate isotopic response and bound fractionation or unequal enrichment within a source before using Eq. 14; report propagated uncertainty rather than forcing fractions into the interval zero to one.

Collect gas, condensates, soluble and insoluble residue, and catalyst-bound carbon. Consistent ethylene enrichment classifies retained carbon from earlier charges by its original source, but does not identify which PE charge produced a later molecule. Use cumulative sequence output for the main decision. Account for beginning and ending retained inventories; count collected products once and do not also credit intermediate residue disappearance as recovered product. Charge-resolved kinetics remain useful, with carryover stated explicitly. A polymer-free blank bounds background reactions but does not reproduce a polymer-conditioned catalyst.

3.3 Experiment 1: reproduce reuse and bracket the fresh-Na requirement

Use one characterized clean HDPE lot and reproduce the catalyst preparation, one fresh time course, and the three-charge unsupplemented and once-supplemented sequences. The original sealed procedure charges ethylene and methane before heating; its nominal 15 bar ethylene is not a controlled 15 bar hot partial pressure. Record fill temperature, gas quantities, pressure convention, actual hot pressure, and temperature history. The separate-component figure also lists 10 bar where the associated text and supporting procedure specify 15 bar; use the detailed method and retain the discrepancy in the replication record.[15, 21]

The finite dose bracket is Table 4. The 0.2 g addition is an analyst-selected midpoint. All reused arms add fresh Na material before charge 2 only; none receives another addition before charge 3. Retain the entire relevant mixture without washing or selective catalyst separation. Use independent sequence replicates, blocked by preparation and run order. The all-fresh arm consists of three fresh catalyst charges and supplies the output reference for material accounting.

Table 4: Fresh catalyst solids consumed over three 1 g PE charges. The lower dose is proposed; the full once-only dose is the published comparator.

Policy	Na/alumina (g)	W/silica (g)	Total (g)
Fresh mixture for every charge	1.2	1.2	2.4
Retained, no makeup	0.4	0.4	0.8
Retained, 0.2 g before charge 2	0.6	0.4	1.0
Retained, 0.4 g before charge 2	0.8	0.4	1.2

If $f = Q_{\text{PE, reuse}}/Q_{\text{PE, allfresh}}$, the published once-supplemented policy gives a total-solid productivity ratio $2f$ relative to all-fresh operation. The midpoint policy gives $2.4f$. At equal output, the midpoint would consume 25% less fresh Na/alumina and 16.7% less total catalyst solid than the full supplement. These are conditional arithmetic results, not measured savings. Three doses bracket demand; they do not locate a precise optimum.

The reported restart cools and dismantles the reactor under inert conditions. Include fresh catalyst subjected to a polymer-free thermal/handling cycle before its first PE charge, to determine whether interrupted operation alone changes function. It is a handling/conditioning control, not an exact reproduction of a spent polymer environment. Record solid recovery, retained carbon, exposure duration, and transfer losses. If existing hardware permits closed inert recharging, compare it only after validating delivered PE mass, temperature history, and mixing. Do not presume that avoiding opening changes nothing else.

One added-support control, using the same dried support and the full supplement’s solid mass, tests whether increased solids/contact explains part of rescue. It cannot duplicate Na/alumina surface chemistry or rheology. Measure a limited agitation/particle-size sensitivity where the existing

apparatus permits it; a stirring plateau does not exclude pore access limitations. If handling dominates the loss, develop a useful handling protocol at that scope rather than calling the decline intrinsic turnover-driven deactivation.

Before decisive runs, fix accepted cumulative polymer-carbon output, allowed product distribution, complete-time ceiling, and the smallest material saving worth detecting. Set balance tolerances and independent replication from pilot uncertainty so that the desired saving is resolvable. Report each charge’s output as well as the cumulative value, preventing a policy from concealing a stalled final charge behind early production.

3.4 Experiment 2: one lower ethylene condition in the same reactor mode

The pressure intervention requires a semibatch reactor with continuous gas flow and product removal. The published reuse benchmark is sealed batch, so first establish fresh/retained differences and the direction of Na rescue in the semibatch mode actually used. If that mode does not show a consequential reuse deficit, close the pressure-as-rescue branch; changing reactor mode has already changed the question.

Choose one lower ethylene partial pressure against a qualified reference, using inert replacement at fixed total pressure, inlet molar flow, temperature, PE charge, catalyst inventory, and agitation. As a concrete initial contrast, halve the reference inlet ethylene fraction while retaining a constant carbon-free tracer fraction and replacing the balance with Ar. This is a proposed diagnostic amplitude, to be accepted only within the qualified gas-delivery range. Measure reactor/outlet composition and actual molar flow rather than assigning reactor ethylene activity from the cylinder blend. Document absolute versus gauge pressure. The source semibatch procedure’s 19 bar gauge total pressure is not interchangeable with its sealed-batch ethylene fill pressure.[21]

Use complete-charge comparisons on independently conditioned catalyst inventories. Match all pre-intervention history through charge 1 at the reference condition, then randomize retained inventories across both pressures and the existing 0, 0.2, and 0.4 g Na/alumina doses: six independently replicated conditions. Each addition occurs once, before charge 2; none occurs before charge 3. Test the nonzero doses regardless of the zero-dose result, because pressure and makeup can interact. Include fresh mixtures at both conditions to establish whether the response is specific to retention. This is an operating intervention on a changing polymer/catalyst mixture, not an isolated intrinsic reaction order. The source itself improved semibatch output while increasing both pressure and flow; that result cannot establish a pressure order, but it is substantial evidence against assuming that less ethylene must help.[15]

Do not begin with rapid pressure waveforms. The published apparatus samples every 12 minutes, and gas holdup plus dissolved-product release may obscure shorter responses.[21] A reference–lower–reference transient is useful only if measured gas and product response times are short compared with chain evolution and catalyst change. Otherwise, complete-charge output gives the defensible comparison. A temporary propylene spike caused by release of stored product is not restored catalytic production.

A useful reduced-dose policy must satisfy polymer-carbon attribution, full product collection, product specification, and the fixed complete-time ceiling. Compare each reduced dose at both pressures against the full-dose reference-pressure policy, reporting pressure contrasts within each dose with uncertainty. A zero-dose null closes only pressure rescue without makeup; the broader tested saving claim requires informative results at the reduced nonzero dose as well. Test fixed lower ethylene from the start of all three charges if the fresh comparison supports it; a simple fixed condition is the

preferred operating rule when it performs as well as a history-dependent switch. Count inert supply, separation, and circulation. Lower ethylene partial pressure at unchanged total pressure does not demonstrate lower compression demand or lower stoichiometric ethylene consumption.

The adverse outcome is equally plausible. Ethylene participates in metathesis, may accept hydrogen during initiation, and changes swelling and gas/polymer transport. Lowering it can slow one function more than it relieves another, change selectivity, or reduce contact. Close the tested partial-substitution branch only if the crossed comparisons exclude a consequential saving that meets the output, specification, and time requirements with informative uncertainty. This conclusion covers the measured doses and two pressures, not every unmeasured intermediate dose. It does not identify which catalyst function decayed, and it does not justify adding a pressure library until an apparent improvement appears.

3.5 Chain-state evidence and the limit of a selective-initiation claim

Track product trajectories together with full molecular-weight distributions, recovered fractions, and unsaturation where measurable. Endpoint olefin content reflects formation and consumption simultaneously; low unsaturation does not prove slow initiation. Nor can a bulk C=C fraction establish how many chains contain reactive sites. Heating alone changes molecular weight in the source’s component experiments, further limiting mechanistic assignment from a mean molecular weight.[15]

Qualify the quench before treating recovered chains as the reacting state. Wang’s dissertation reports that depressurization and two to three hours of cooling permitted isomerization and fragment recombination in its tandem system.[20, ch. 4, pp. 72–74] On Na/W, compare the fastest technically validated recovery with a deliberately delayed recovery at the same conversion window, accounting for all fractions. If unsaturation or chain distributions depend materially on workup, report them as post-recovery descriptors and close direct hot-state inference. Solvent-soluble material alone cannot represent carbon retained on inaccessible solids.

If the pressure intervention gives a real saving, select one explanation for a further discriminating test. A matched unsaturated PE and hydrogenated counterpart can test whether feed state changes the pressure benefit, provided chain-length/branching distributions, workup residues, and metal contamination are controlled. The prediction must come from the measured Na/W response, not from transferring the molecular Ru model’s optimum. This is a conditional mechanistic extension, not a second mandatory first campaign. Different unsaturation also changes the reacting chain distribution and contact, so feed rescue alone does not isolate initiation.

A component-specific claim requires operationally selective probes whose specificity is demonstrated on the separate catalysts and supports, and whose access and conditioning effects are assessed. Small olefins may reach sites inaccessible to polymer or activate the material being measured. If those constraints cannot be met, retain an empirical relation among history, pressure, and fresh-material demand. The useful operating result does not require claiming that initiation alone decays while downstream functions remain intact.

3.6 Experiment 3: predict a catalyst-use decision, then test whether it persists

Use the crossed three-charge dose brackets and complete time courses to construct the simplest predictor of accepted output at both pressure conditions, retaining any pressure–dose interaction. Start with direct interpolation over the measured makeup range and measured uncertainty; do not

fit separate unobserved initiation and metathesis capacities to the same product trace. If output is not sufficiently regular with dose to support interpolation, select among tested policies rather than claim an unmeasured optimum.

Withhold one intermediate makeup dose from calibration. Choose it prospectively only if the predictor says it can meet the accepted output and time criteria with a resolvable saving; then validate it at *both* pressures on independent three-charge sequences with matched charge-1 histories and the dose supplied before charge 2 only. Freeze both predictions before collecting either result. Compare against the best tested fixed-pressure policy and the published full-supplement schedule in the same reactor mode. Successful validation means that the lower confidence bound on accepted output meets the predeclared requirement, product specification and time are met, and fresh-material demand is lower after attribution uncertainty. To claim that pressure enables this dose saving, require the lower-pressure policy to pass and the same dose at reference pressure to fail an acceptance criterion with informative uncertainty. If both pass, report an adequate lower dose and any separately resolved pressure benefit, without claiming pressure was necessary for the material saving. An unresolved reference result leaves that necessity unestablished. No minimum over unmeasured doses is identified. A failed prediction remains a failure; do not refit it and call it validation.

This first result establishes a bounded inventory saving. To test whether it merely defers catalyst replacement, extend the selected policy and comparator through further 1 g PE charges. Before the extension, freeze a per-charge intervention trigger based on accepted output, quality, and the charge-time ceiling, a common cumulative accepted polymer-carbon target, and maximum complete-time and charge-count bounds. At the trigger, collect products and remove the entire retained catalyst/residue mixture under the qualified handling protocol; quantify its solids and unprocessed carbon. Replace it with 0.4 g fresh Na/alumina and 0.4 g fresh W/silica, then restart the assigned validated pressure schedule and once-only makeup before the new mixture’s second charge. Preserve the common reference-condition first charge for policies validated with that history. Do not attempt selective W replacement or unqualified regeneration.

Continue until the common output target is reached or either maximum bound is reached. Charge all fresh solids and complete time, including failed charges, cooling, removal, recharge, reheating, collection, and final inventory closure. Withdrawn and final residues remain in the carbon and waste accounts; unprocessed carbon is not recovered product. Use the common target as the credited output for the target comparison and report any collected excess separately. A policy that fails to reach the target receives no extrapolated credit. This tests sustained demand only over the bounded extension with the specified full-mixture replacement rule; any shorter service interval must pay its measured replacement cost.

The scientific payoff grows when a measured chain or catalyst-state descriptor predicts a new history better than charge count alone and thereby changes a real makeup decision. Without that link, a reproducible fixed-pressure saving remains a useful bounded operating improvement. If no qualified intervention reduces demand at accepted output and time, close this proposal’s improvement claim. Neither a negative result nor successful fresh-Na rescue establishes a unique molecular cause of reuse loss.

4 When does nickel improve the useful life of promoted silver?

4.1 The opportunity and the claim that survives prior work

Ethylene epoxidation must favor ethylene oxide (EO) over combustion while maintaining enough reaction rate to produce EO economically. Selectivity alone can reward an almost inactive catalyst. The proposed program therefore asks whether dilute nickel changes the sustained output of a functioning Cs/Re/chloride-promoted Ag catalyst enough to justify a different material or operating policy. It starts with ordinary operation, then tests one prediction about the observed difference. It does not presume that Ni preserves a particular oxygen species or Ag–Re arrangement.

The potential value can be illustrated without predicting a catalyst improvement. If EO and complete combustion are the only carbon outlets, making one mole of EO at fractional ethylene selectivity S consumes $1/S$ moles of ethylene and $0.5 + 3(1/S - 1)$ moles of oxygen, and produces $2(1/S - 1)$ moles of CO_2 . Raising S from 0.90 to 0.91 at fixed EO output would reduce these reaction-level requirements by 1.10% for ethylene, 4.40% for oxygen, and 10.99% for CO_2 formation. Recycle, separation, utilities and catalyst manufacture are outside this calculation. A useful measured improvement must also survive those costs.

Ni-dependent retention on a fully promoted catalyst is already reported. Kemp’s 1995 patent describes Ni added after preparation of an Ag/Cs/Re/S catalyst, with lithium-containing precursors also used. Its Ni-containing catalyst lost 4.3 selectivity percentage points after 50 days of accelerated aging, compared with 6.1 points for its Ni-free parent; measurements followed return to 40% oxygen conversion and reoptimization. Both started within a reported 86.0–86.5% selectivity range. The bulk Ni:Ag atomic ratio was approximately 1:835. However, the aged pair differed in Cs content and post-impregnation treatment. A solvent-treated, Cs-matched Ni-free control had poorer initial performance and was not aged. Thus the disclosure establishes relevant prior performance, while leaving the isolated Ni effect and useful integrated output unresolved. The endpoint difference is a difference in losses from separate starting values, not an exactly measured final-selectivity difference or a demonstrated lifetime extension.[22]

Jalil and colleagues provide a different starting point: dilute Ni suppresses combustion more strongly than EO formation on supported Ag, and organochloride gives a further selectivity increase. Their main materials lack the complete Cs/Re promoter system. Their processing controls, model-surface experiments and Ni spectroscopy support a Ni-associated effect, but nominal loading does not count working surface Ni atoms. The supplementary chloride-response trace shows an initial EO-rate increase followed by substantial decline on NiAg. It demonstrates evolving cooperation, not a stationary, durable output advantage on the proposed formulation.[23, 24]

Gas-phase alkyl-chloride modifiers deposit chlorine on Ag, while competing processes, including removal by alkanes such as ethane, reduce the retained chlorine. This balance changes oxidation rates and selectivity, so chloride adjustment can trade selectivity against useful EO rate. Its optimum depends on the material and operating conditions; neither a universal coverage nor a universal modifier setting follows.[25]

Hwang and colleagues link the distinctive Cs/Re/chloride response to promoter architecture and propose conversion of otherwise unselective oxygen into an additional epoxidizing intermediate. That proposed sequence motivates a compatibility question; it is not direct evidence that Ni supplies oxygen to Re or preserves their contact.[26] The competing operating explanation is equally serious. Iyer and Bhan already combined independently assessed oxidation, chlorine exchange and support-mediated EO degradation in a forward reactor prediction. Santos and colleagues demonstrate

that temperature, oxygen, water and CO₂ change the chloride response.[25, 27] Lower-chloride aging policies and materials with a more stable optimum chloride requirement also appear in prior disclosures. The former includes 28-day experiments followed by longer-term extrapolation; the latter maintains work rate by adjusting temperature. Neither a fitted distant crossover nor a stable chloride optimum proves stable activity.[28, 29]

The original contribution sought here is therefore narrow: a composition- and processing-controlled, Ni-dependent relationship that predicts when changing chloride operation can substitute for Ni, or when a useful Ni advantage survives that alternative. Better controls alone do not establish that contribution. A practical benefit may be established without a unique microscopic explanation; a mechanistic claim requires a discriminator beyond a selectivity curve.

4.2 Stage A: qualify one material pair and a fair output comparison

Preserve the functioning promoter architecture. Use an existing, reproducible Cs/Re-promoted Ag/alpha-alumina reference and its qualified epoxidation reactor. Access to that platform is an entry requirement. Transferring every promoter to a dilute colloidal-Ag support could remove the reference function before Ni is tested. Hwang’s Ag/support area dependence makes preserving the useful reference architecture more defensible than matching nominal particle diameter alone.[26]

Make three materials from each parent preparation: untreated reference, sham-processed reference, and Ni-treated reference. A first, historically grounded route is post-impregnation with the nickel-nitrate/amine-containing solution and staged drying described by Kemp, adapted to the existing material only after a small compatibility check. Start near a bulk Ni:Ag atomic ratio of 1:800, an illustrative preparation choice close to the historical dilute loading, rather than assuming Jalil’s 1:200 optimum transfers. Apply all solvent, ligand, drying and handling steps to the sham, omitting the Ni precursor and documenting any unavoidable solution-composition difference. Measure actual Ag, Ni, Cs and Re contents, other intentional promoters, and Ag size distributions before claiming a matched pair. Matching gross inventories does not establish identical promoter placement.

The untreated reference diagnoses damage caused by the added processing and remains an ordinary-aging policy comparator; the sham is the causal comparator for the complete Ni-addition intervention. If the sham loses the reference’s useful response, permit one correction tied to an identified preparation problem, then stop this route if the problem remains. Fresh qualification cannot exclude a processing penalty that emerges later, so untreated operation must also be followed over the complete comparison horizon. A Ni/sham retention difference without a useful advantage over untreated operation is a processing-dependent result, not an improvement of the starting catalyst. The historical route does not establish isolated Ni in Ag. Add an unpromoted Ag/Ni–Ag bridge only if attributing the result specifically to Jalil’s fresh-chemistry function; a full Ni-by-Re composition series is a later discriminator, not an entry requirement.

Separate common conditions from useful operating choices. Commission gas balances, temperature uniformity, transport sensitivity and feed-to-analyzer response before assessing small selectivity differences. Measure EO, ethylene, oxygen, CO₂, water and any significant carbon byproducts, with an independently calibrated flow tracer. Distinguish net formation from outlet concentration when CO₂ or EO is cofed. Ethane used to remove chlorine is another possible carbon source: measure its conversion and do not assign all CO₂ to ethylene combustion. The ratio $r_{EO}/(r_{EO} + r_{CO_2}/2)$ is an ethylene selectivity only when its carbon-source and product assumptions hold. Use consumed ethylene and a closed product balance for the reported selectivity; unresolved

cofeed contributions limit interpretation.

Apply a common initial conditioning protocol to untreated, Ni and sham materials, and record their resulting states instead of forcing equal initial selectivity by different treatments. Matching a reproducible history is feasible; identical microscopic starting states are not guaranteed. Use a small overlapping bracket of chloride settings with reference returns, keeping temperature, pressure, ethylene, oxygen, ethane and flow common for the Ni/sham causal comparison. Qualify a useful untreated-reference setting within the same operating and tuning bounds. Specify absolute pressure and the actual chloride feed, not only a nominal chloriding index. Wait according to observed rates and gas holdup, with a bounded maximum dwell. If reference returns remain history dependent, report age-specific trajectories rather than assembling different ages into a fictitious steady curve. Independent preparations of the decisive contrast are required; repeated samples from one bed do not measure synthesis variation.

Use these data for two distinct questions. At overlapping conditions, does Ni change the EO and combustion rates? Under each material’s own selected chloride setting, which choice gives more accepted EO? A shifted chloride optimum can answer the second without proving altered oxygen chemistry. Do not require a fresh Ni advantage to proceed: if both materials qualify, a fresh null still permits a later retention difference.

Define the output requirements before the ordinary-operation comparison: an accepted ethylene-conversion interval, a selectivity floor, a minimum EO work rate, and common temperature and feed limits. Set these from the reproduced reference and the intended operating decision, not from whichever endpoint favors Ni. Fix a finite complete observation horizon H , starting with initial conditioning, and the smallest useful contrast using commissioning uncertainty and available run time. An unresolvable contrast requires a better measurement or a narrower claim, not an assumed long-term advantage.

For policy p , use the complete elapsed-time productivity

$$P_p(H) = \frac{Q_p(H)}{m_{\text{Ag},0}H}, \quad Q_p(H) = \int_0^H \dot{n}_{\text{EO},p}^{\text{accepted}}(t) dt. \quad (15)$$

The accepted rate is zero for unqualified product or downtime; product-acceptance windows must be fixed and compatible with analyzer response. Separately predeclare production windows and a return-assessment window in which conversion, selectivity and work-rate requirements apply. Planned conditioning, recovery and return intervals may have low or zero output, but their maximum durations and return deadline are fixed before the run. All elapsed time and resource use remain charged, and product criteria apply to every credited EO amount. Failure during a required production window or failure to return within the allowed interval makes a policy inadmissible; an inconvenient low-output interval cannot be relabeled afterward as planned recovery. Temperature and feed bounds apply throughout.

Report the unfiltered net EO integral and the timing of every qualification failure alongside Q_p . Policy inadmissibility does not make chemically usable EO disappear; distinguish it from actual off-spec product and retain all physical flows in the material balance. Include initial conditioning, chloride searches that recur in the proposed policy, recovery and return to production. Also report output per initial total catalyst mass and occupied bed volume, since Ag normalization alone can hide dilution or additional material. Integrate ethylene and oxygen consumption, net CO_2 generation, chloride and ethane use, and temperature history over the same clock. Higher selectivity with lower work rate or higher required temperature remains a tradeoff.

For each admissible candidate policy p versus a named tested comparator b , state a preparation-cost

boundary per bed campaign over H , with matched initial Ag inventory and actual bed masses and volumes reported:

$$C_{\text{prep},p} - C_{\text{prep},b} \leq v_{\text{EO}} [Q_p(H) - Q_b(H)] - \sum_k v_k [U_{k,p}(H) - U_{k,b}(H)] - v_L [L_p(H) - L_b(H)]. \quad (16)$$

Here v_{EO} is value per mole of accepted EO, U_k are resource or disposal quantities with explicit unit costs v_k , and L is operating handling labor with cost v_L per labor-hour. Include measured feed and modifier demands, utilities with a stated measurement or estimation basis, and any valued emissions or waste. Preparation costs include the initial material and its processing; do not count those costs again as operating resources. Downtime is already charged through complete-clock output and is not automatically an extra labor cost. Without defensible valuations, report this symbolic boundary and the physical differences, not a numerical monetary saving or economic optimum.

4.3 Stage B: ordinary drift before an excursion campaign

After the exploratory measurements, assign independent charges within each preparation block to the Ni/sham common-setting pair and the useful policies identified above, including at least one independently replicated untreated-reference policy. Start their comparison clocks with the same conditioning protocol and follow all arms over the same elapsed horizon. A common-setting arm can also serve as a policy arm when its entire history is identical; otherwise use separate beds or independent matched-age runs. Randomize reactor positions or run order within preparation blocks, include reference checks for instrument drift, and establish uncertainty from independent runs. The common-setting pair identifies the effect of the Ni intervention under that exposure; separately chosen policies compare what an operator could use. They are not interchangeable estimands.

Select the untreated policy from its own measured chloride response: use one lower-chloride setting if it meets the requirements, otherwise a credible reference setting or scheduled chloride adjustment fixed before aging. Do not copy a fitted percentage of another catalyst’s optimum. Ni and sham receive the same operating bounds and opportunity to select a useful policy, with equal limits on tuning effort. Hold the other operating variables common where feasible. If maintaining accepted conversion requires different flow or temperature, that change belongs to the policy, and its rate, residence-time, product-exposure and temperature consequences enter the comparison. At least one credible untreated policy is indispensable even if its fresh response matches the sham. It need not receive the full chloride or recovery factorial. Report the best tested untreated policy explicitly in absolute output/resource rankings.

Compare cumulative EO and reactant loss throughout H , not just endpoint selectivity. Lower-chloride operation may pay an ongoing selectivity penalty; it cannot be booked as a one-time startup cost. An unchanged optimum does not imply that the rate or peak selectivity is unchanged. Likewise, a temperature controller can conceal activity loss unless its full trajectory is retained. No accelerating exposure is needed to create the primary result. If neither ordinary drift nor a useful material/policy contrast is resolved within H , close the program at that preparation and operating scope. An arbitrary severe chloride pulse or an extrapolated crossover cannot supply the missing practical burden.

If a consequential loss develops, compare one mild recovery intervention with ordinary continuation on matched-age beds of both materials, using their prespecified common-setting aging history. Choose the recovery chloride setting from the established useful bracket, hold other feeds fixed, and return both beds to the common reference setting for a prespecified observation window. This is a test of recoverable conditioning under that intervention, not a general regeneration claim. A failed

recovery establishes persistence over its tested duration; it does not establish absolute irreversibility. The recovery and continuation arms must retain separate subsequent trajectories because treatment may alter later drift.

4.4 Stage C: distinguish useful memory from familiar chlorine response

The first explanatory baseline is the measured chlorine/product response on each material, informed by established chlorine-exchange and reactor models.[25, 27] Calibrate it on the actual formulation and feed domain, with independent checks of response times and, where conversion requires it, axial product gradients. Lower combustion generates less water and CO₂ within the bed, so equal inlet products need not give equal local states. A transferred literature model with uncalibrated parameters is not a fair null model for new chemistry.

Compare the EO–CO₂ rate trajectory during recovery with independently measured steady or matched-age chloride-response curves. A trajectory following those curves can still have a useful Ni-dependent recovery speed; curve collapse does not predict the time taken to move along them. Conversely, departure from a whole-bed curve can arise from axial mixtures of chlorine states, temperature changes or product gradients. It does not identify promoter relocation. Add a slower functional state only if a calibrated chlorine/product baseline fails by more than its prediction uncertainty and a consequential unexplained difference remains. The state then denotes reproducible rate memory, not a counted population of selective oxygen sites.

Use at most one additional discriminator selected by the result. If reduced self-generated CO₂ appears to explain the Ni benefit, compare the pair under a common, externally controlled product burden and predict an untested burden after accounting for bed gradients. If ordinary loss survives reconditioning and an inventory change is plausible, examine retained elemental amounts and Ag morphology on sister samples. Unchanged total Re cannot exclude redistribution; a changed mean particle size need not explain all lost function. Ordinary EO cofeed measures net product sensitivity because it can change both EO formation and destruction. Contact-time curvature is also insufficient to identify a secondary combustion flux.

A working-state assay is justified only when it resolves the surviving alternatives. Chlorine stripping is an operational inventory with desorption and normalization assumptions. Trifluoroethanol titration is destructive and estimator dependent: Iyer and Bhan explicitly treat total uptake as an upper-bound site estimate because inactive adsorption is possible.[30] Neither becomes an exact active-site count on Ni-containing material without new validation. Use sister samples, qualify changes during quench and transfer, and require enough sensitivity to detect the minority population invoked by the explanation. An unchanged element-average spectrum cannot exclude such a minority when its abundance is unconstrained.

Oxygen-isotope return measurements and a Ni-by-Re composition series remain optional extensions. They are not needed to decide which tested policy produces more EO. A return experiment would additionally require a justified relation between the labeled population and oxygen actually returning to gas, a complete balance and a resolvable signal. Neither high selectivity nor failure to detect gas return alone proves direct Ni-to-Re oxygen transfer.

4.5 Stage D: predict one operating decision and its cost

The principal follow-on, if ordinary drift proves partly recoverable, is a prediction of when a bounded chloride recovery pays for itself. For both Ni and sham, use the matched-age recovery and

continuation traces to calibrate the simplest time-dependent description consistent with both EO and CO₂ rates. A single relaxation time is permissible only if its approach and return are resolved and repeatable; an unobserved asymptote and relaxation time cannot both be estimated reliably from a short early segment. Use a second training duration to check the description before selecting a withheld duration within the tested range. Rates alone need not uniquely identify surface coverage, and the prediction must not rely on that identification.

At a fixed common remaining horizon H_r , define the recovery benefit relative to continuation as

$$B_m(t_R; H_r) = \int_0^{H_r} [\dot{n}_{\text{EO},m,R}^{\text{accepted}}(t; t_R) - \dot{n}_{\text{EO},m,C}^{\text{accepted}}(t)] dt, \quad (17)$$

where m specifies Ni or sham, t_R is the recovery duration, and both trajectories start at the same defined operating age a , leaving $H_r = H - a$. The integration includes recovery itself and all return time; recovered peak activity is not the benefit. Evaluate resource differences on this same boundary. The model must predict the sign and an uncertainty interval for B_m and the EO/CO₂ trajectories for both materials. Construct each recovery policy's absolute campaign output as

$$Q_{m,R}(H) = Q_{m,C}(H) + B_m(t_R; H - a), \quad m \in \{\text{Ni, sham}\}, \quad (18)$$

using its own continuation baseline with the identical pre-intervention history, initial clock and inventory basis. Compare *all admissible tested* Ni and sham ordinary and recovery policies, plus untreated ordinary operation, at the same H , with the same acceptance rules and full resource accounting. Report inventory-normalized outputs and the preparation-cost boundary on the stated matched-inventory basis. A sham recovery beating ordinary Ni does not remove a useful Ni advantage if the tested Ni recovery policy is better; a larger within-material benefit also does not determine the best absolute output. A comparison against only ordinary Ni supports that narrower conclusion.

Recovery learned under the common-setting history cannot be assumed to transfer to a different policy history. Lock the withheld duration, return setting, acceptance windows and predictions for both Ni and sham before testing their recovery and matched continuation on independent preparation blocks. Validate both absolute recovery outputs and their decisive ranking, with uncertainty, against each other and the tested ordinary policies, including untreated operation. Do not retune chloride after observing the withheld result, and do not exclude a tested Ni recovery policy simply because sham recovery has already beaten ordinary Ni. This ranks the finite tested policies, not an unmeasured global optimum.

Success requires an observed decision-changing benefit or limit within H_r , with predictive accuracy sufficient to resolve the predefined useful contrast. A familiar chlorine model that already predicts the result supports an operating choice but closes the claim of an additional Ni-specific mechanism. If a Ni-dependent memory term is necessary, it must improve withheld prediction beyond that calibrated baseline and identify a useful preparation or operating dependence. One successful recovery does not justify a periodic-service model: repeated returns to the same productive state and measured recurrence costs are prerequisites for that extension.

If the important ordinary difference is persistent rather than recoverable, replace this branch with one prespecified product burden or exposure duration that the surviving explanation predicts will retain or remove the Ni advantage. Do not pursue all branches in parallel. A result that only reproduces historical Ni retention, that disappears under familiar chloride adjustment without a further consequential relationship, or that requires an open-ended synthesis campaign should remain a bounded negative or confirmatory study. A successful program would instead deliver a reproducible formulation/policy comparison and a validated prediction of where its advantage ends. Competitive industrial lifetime remains outside the evidence of a finite laboratory campaign.

5 Other considered ideas and claims not advanced

These six ideas are ranked for possible reconsideration, rather than proposed as six additional full campaigns. Each has a specific unresolved decision. Existing equipment may change the order in which a small test is worthwhile.

1. Tungsten coordination and retention in sugar conversion: exploratory hold. The opportunity is to connect productive sugar coordination, W export and useful glycol output to a controllable anchoring or feed variable. Ti–O–W stabilization and reuse already have strong precedent. Li’s supported-Ni control without intentionally added W is labeled Ni/M. Its 29.5% ethylene-glycol yield with preleachate must be compared with 27.1% for Ni/M alone; it is not a 29.5% leachate-derived contribution.[31, 32] If reopened, first reproduce a retained and a soluble-W reference with hot export and carbon balances. Transferred activity can demonstrate exported function but cannot uniquely assign all upstream turnover. A chemically grounded intervention addressing a consequential reference limitation remains missing.

2. Product-rich liquid Ti-zeolite epoxidation: conditional model-transfer test. Ask whether a calibrated reaction network predicts an independent product-accumulation trajectory, including peroxide loss and epoxide ring opening. Product-induced solvent changes already have direct spectroscopic precedent, and related Ti-zeolite secondary kinetics contain unresolved behavior.[33, 34] Qualify component kinetics in the chosen liquid catalyst/solvent system before freezing the prediction; a vapor-fed model is not a validated liquid baseline. Close the additional-mechanism claim if the established network predicts useful output adequately. A residual alone does not identify solvent reorganization, fouling or Ti loss.

3. Zirconia styrene self-cleaning: bounded chemical-balance pilot. Hydrocarbon-assisted removal of inhibiting water is already proposed for Zr–O acid–base pairs; the cleaning products were not identified in the source’s dilute-feed regime.[35] First establish useful net output with measured moisture and relevant styrene/hydrogen burdens, then cross water history with ethylbenzene exposure to relate oxygen export to recovered function. Control oxygen exchange and perturbation by the activity assay before isotope attribution. Expand only if the chemistry predicts a useful feed or removal policy; omitting steam alone does not establish process savings.

4. Does an acid-site assay change subsequent zeolite aging? Conditional protocol study. Compare uninterrupted aging of protonic chabazite (H-CHA) with repeated NH_3 /water assays and a matched thermal/water sham. Measure function before terminal titration on separate matched specimens; a common terminal assay then compares recoverable capacity. Repeated assays are part of the relevant published protocol; a Cu/SSZ-13 aging-order comparison also assayed both arms and therefore does not answer assay versus no assay.[36, 37] The useful result would predict later function under a withheld assay spacing. Immediate uptake recovery is insufficient, and a null schedule effect cannot by itself determine the lifetime of an unseen Al intermediate.

5. Wet methane oxidation after NO removal: application-dependent diagnostic. Test whether the sulfur-tolerance advantage of Pd/SSZ-13 treated by ST-CO-N₂-O₂ over its ST-CO-O₂ counterpart survives removing NO at matched water, sulfur history, temperature and precious-metal inventory. The former includes an extra N₂ treatment between CO and O₂. The reported sulfur

interval includes NO, while history-dependent NO promotion on other Pd catalysts is already known.[38, 39] Apply the small crossed NO/sulfur test to both variants only where a real low-NO, sulfur-containing duty could change the material choice. Without that application boundary, another NO response is a diagnostic rather than a substantial program.

6. Ga-based propane dehydrogenation regeneration: conditional cycle study. Test whether a measured restoration endpoint predicts a shorter useful regeneration across a withheld spent history. Redox-related deactivation and shortened oxygen treatments are already reported.[40, 41] Oxygen consumption minus CO₂ alone is not a selective Ga-redox inventory: oxidation of retained hydrogen and other oxygen-containing products also enters the balance. Calibrate that balance before assigning sites, and compare complete-cycle propylene output, gas use and heat duty with the best simple regeneration policy. Correcting an assay does not by itself establish a better catalyst cycle.

Claims withdrawn, rather than topics declared invalid

Several attractive shortcuts do not survive scrutiny. A faster outlet-water transient does not uniquely establish lower active-site exposure. Fresh-Na rescue does not identify selective initiation loss. Ni compatibility and retention on promoted Ag are already prior art. The detailed chapters replace these claims with narrower functional and predictive questions; none of these corrections invalidates its entire catalytic field.

For tungsten, the inactive-mannose-reservoir premise should also be withdrawn: published hot mannose feeds produce glycolaldehyde, with epimerization complicating assignment of the direct cleavage route.[42] This does not exclude an inactive fraction, but mannose cannot be assumed to form an inactive reservoir. The proposal remains on hold until a better supported intervention is selected. More broadly, the shortlist distinguishes missing evidence from a demonstrated contradiction: an incomplete program can be reconsidered after its specific missing test, whereas an invalid inference should not be retained as the default mechanism.

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